

123456

SIMPLE HOBBY CONTROLLER – TWO READABLE SCHEMATIC PAGES

Power + dual wheel driver

File: simple_01_power_motors.kicad_sch

ESP32 + servos + UART

File: simple_02_esp_servos_uart.kicad_sch

Harrison

Sheet: /

File: pickleball_robot_simple.kicad_sch

Title: **Pickleball Robot – Simple Hobby Controller**

Size: A4

Date: 2026-07-12

Rev: **H1-SIMPLE**

KiCad E.D.A. 10.0.4

Id: 1/3

A

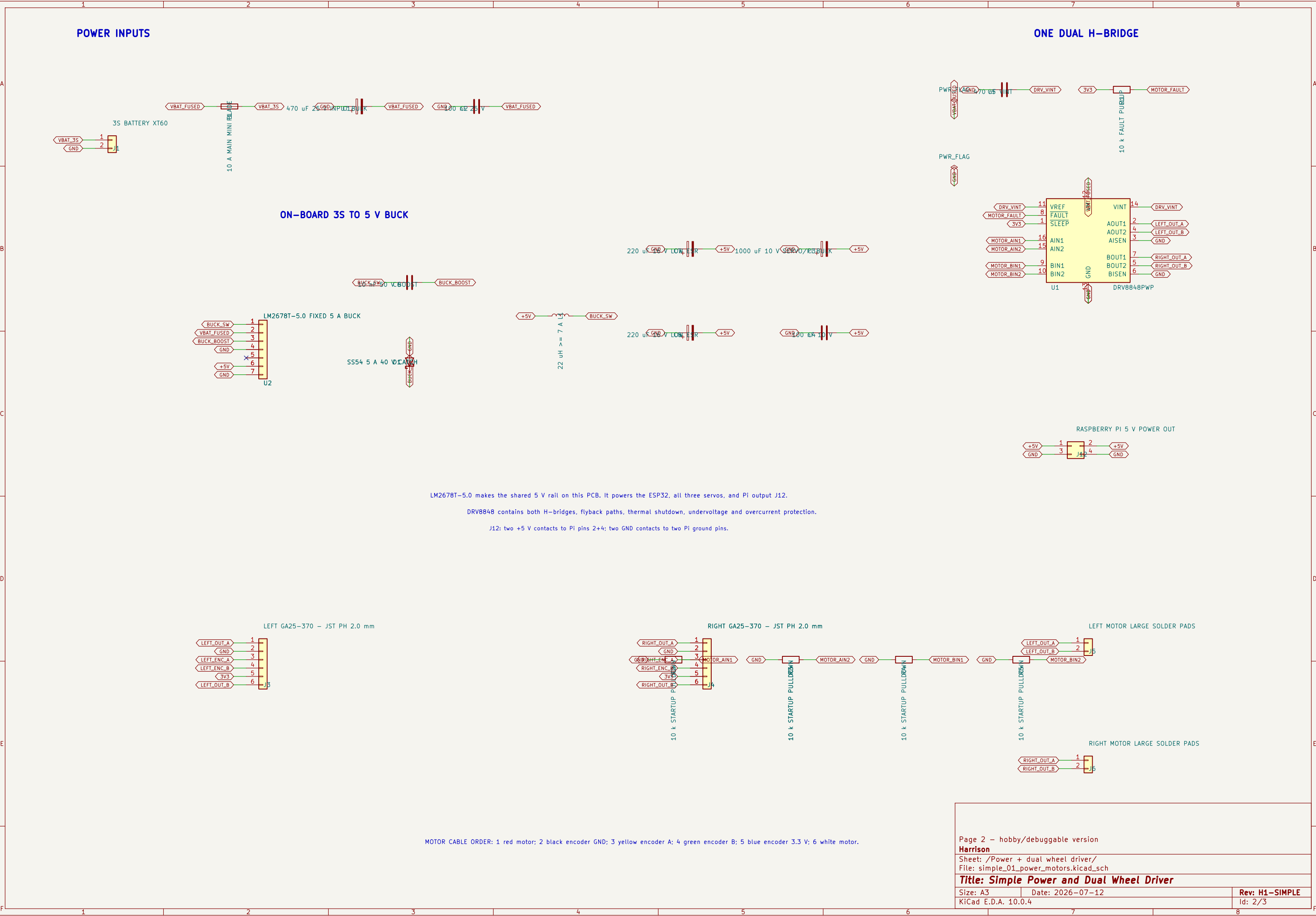
B

C

D

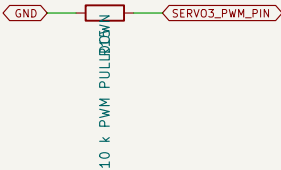
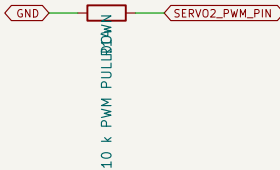
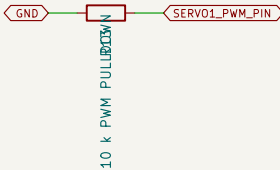
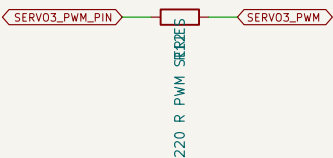
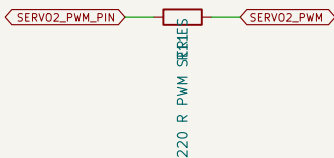
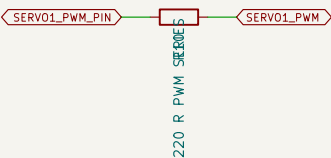
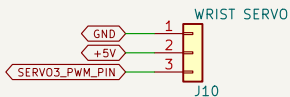
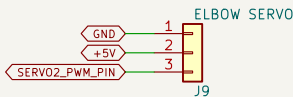
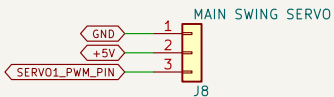
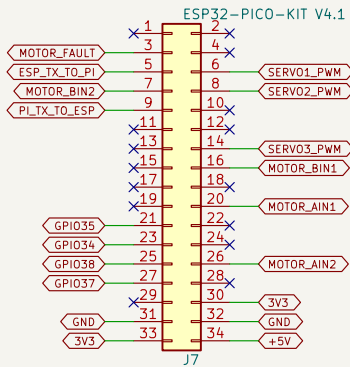
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Open a page from the hierarchy panel. Global labels join nets between the two pages.



ESP32-PICO-KIT V4.1 CARRIER

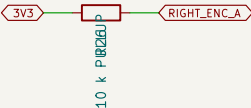
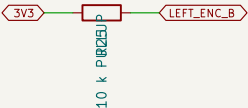
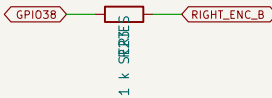
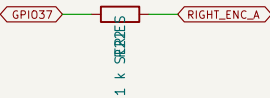
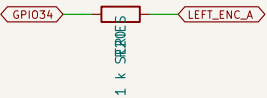
THREE INDEPENDENT 5 V SERVOS



Servo headers: pin 1 GND, pin 2 regulated +5 V, pin 3 separate PWM. Each servo moves independently.

ENCODER INPUTS

RASPBERRY PI 3.3 V UART



J11: 1 GND; 2 PI TX -> ESP GPIO18 RX; 3 ESP GPIO19 TX -> PI RX; 4 is 3.3 V reference only.

Motor encoder outputs are assumed 3.3 V because their blue supply wire is powered from 3.3 V.

J11 is UART only. Raspberry PI power comes from the shared +5 V rail through high-current connector J12 on page 2.

Page 3 - hobby/debuggable version

Harrison

Sheet: /ESP32 + servos + UART/
File: simple_02_esp_servos_uart.kicad_sch

Title: ESP32, Three Servos, Encoders, and PI UART

Size: A3 Date: 2026-07-12

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